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BIPOLAR BATTERY

Abstract

Provided are: plural bipolar electrodes; a separator positioned between a negative electrode active material layer and a positive electrode active material layer which are adjacent to each other; plural primary sealing materials provided at an outer peripheral portion of the current collector, the outer peripheral portion being a region at which the negative electrode active material layer and the positive electrode active material layer are not provided; a secondary sealing material provided at an outer peripheral portion of all of the primary sealing materials; and an electrolytic solution provided at an internal space V, wherein, in a case in which a melting temperature of the primary sealing materials is defined as mp1, a normal battery operating temperature is defined as nth, a boiling point of the electrolytic solution is defined as bp, and a melting temperature of the separator is defined as smp, $\text{nth} < \text{mp1} < \text{bp}$ and $\text{nth} < \text{mp1} < \text{smp}$.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2024-033246 on Mar. 5, 2024 the disclosure of which is incorporated by reference herein.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a bipolar battery.

Related Art

[0003] Japanese Patent Application Laid-Open (JP-A) No. 2019-091606 discloses a bipolar battery. The bipolar battery includes a negative terminal electrode, a positive terminal electrode, plural bipolar electrodes, plural separators, plural primary sealing materials, a secondary sealing material, and an electrolytic solution. Each bipolar electrode is provided between the negative terminal electrode and the positive terminal electrode. Each separator is positioned between an adjacent negative electrode active material layer and positive electrode active material layer which are provided at each electrode. Each primary sealing material is a frame-like member which is provided at an outer peripheral portion of a current collector of each electrode. The secondary sealing material is a frame-like member which is provided at an outer circumferential portion of all of the primary sealing materials.

SUMMARY

[0004] There is room for improvement regarding forced discharge when the temperature is raised, in the bipolar battery of Japanese Patent Application Laid-Open (JP-A) No. 2019-091606.

[0005] In consideration of the above-described circumstances, an object of the present disclosure is to obtain a bipolar battery for which forced discharge performed at the time of temperature rise is difficult to be prevented.

Means for Solving the Problem

[0006] A bipolar battery according to a first aspect includes: plural bipolar electrodes each including a current collector, a negative electrode active material layer provided on one face of the current collector, and a positive electrode active material layer provided on another face of the current collector; a separator positioned between the negative electrode active material layer and the positive electrode active material layer which are adjacent to each other; plural frame-like primary sealing materials provided at an outer peripheral portion of the current collector, the outer peripheral portion being a region at which the negative electrode active material layer and the positive electrode active material layer are not provided; a single frame-like secondary sealing material provided at an outer peripheral portion of all of the primary sealing materials; and an electrolytic solution provided at an internal space surrounded by the current collector, the primary sealing materials, and the secondary sealing material, wherein, in a case in which a melting temperature of the primary sealing materials is defined as $mp1$, a normal battery operating temperature is defined as nth , a boiling point of the electrolytic solution is defined as bp , and a melting temperature of the separator is defined as smp , $nth < mp1 < bp$ and $nth < mp1 < smp$.

[0007] In the bipolar battery according to the first aspect, an electrolytic solution is provided at an internal space surrounded by the current collector, the primary sealing materials, and the secondary sealing material. Therefore, in a case in which the temperature of the bipolar battery is at the normal battery operating temperature nth , the electrolytic solution provided at one internal space

does not flow through a primary seal into an adjacent internal space.

[0008] Further, in the bipolar battery according to the first aspect, in a case in which the melting temperature of the primary sealing materials is defined as $mp1$, the normal battery operating temperature is defined as nth , the boiling point of the electrolytic solution is defined as bp , and the melting temperature of the separator is defined as smp , $nth < mp1 < bp$ and $nth < mp1 < smp$. Therefore, in a case in which the temperature of the bipolar battery is higher than the normal battery operating temperature nth , the primary sealing materials are melted before the temperature of the electrolytic solution reaches the boiling point bp . As a result, in a case in which the temperature of the bipolar battery is higher than the normal battery operating temperature nth , the electrolytic solution provided at one internal space can pass through the region in which the primary sealing materials were present and flow into the adjacent internal space. Therefore, forced discharge occurs between the negative electrode active material layer and the positive electrode active material layer provided at both sides of the current collector by the electrolytic solution having flowed into the adjacent internal space. Namely, the negative electrode active material layer and the positive electrode active material layer provided at both sides of the current collector are equalized in potential.

[0009] Furthermore, in the bipolar battery according to the first aspect, when the temperature of the bipolar battery further rises after forced discharge has occurred between the negative electrode active material layer and the positive electrode active material layer provided at both sides of the current collector, the separator may be melted. However, the negative electrode active material layer and the positive electrode active material layer positioned at both sides of the separator are at equal potentials prior to melting of the separator, such that there is little risk of an excessive increase in the temperature of the bipolar battery.

[0010] A bipolar battery according to a second aspect is the bipolar battery according to the first aspect, wherein bp is higher than $mp1$ by 50°C . or more.

[0011] In the bipolar battery according to the second aspect, the boiling point bp of the electrolytic solution is higher than the melting temperature $mp1$ of the primary sealing materials by 50°C . or more, such that a state in which the temperature of the bipolar battery is higher than or equal to the melting temperature $mp1$ of the primary sealing materials and lower than the boiling point bp of the electrolytic solution is easily maintained for a certain period of time.

[0012] As explained above, the bipolar battery according to the present disclosure has an excellent advantageous effect in that forced discharge performed at the time of temperature increase is difficult to be prevented.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] Exemplary embodiments of the present disclosure will be described in detail based on the following figures, wherein:

[0014] FIG. 1 is a perspective view of a bipolar battery according to an exemplary embodiment.

[0015] FIG. 2 is a schematic cross-section view taken along line 2-2 in FIG. 1.

[0016] FIG. 3 is a cross-section view corresponding to FIG. 2, when a primary seal has been melted.

[0017] FIG. 4 is a cross-section view corresponding to FIG. 2, when a separator has been melted.

DETAILED DESCRIPTION

[0018] FIG. 1 illustrates a bipolar battery **10** (hereafter, referred to as a battery **10**) according to an exemplary embodiment. The battery **10**, serving as a secondary battery, can be installed at various devices. For example, the battery **10** is installed at an electric vehicle (battery electric vehicle (BEV)), and can supply electric power to an electric motor that serves as a drive source. It should be noted that in each of the drawings, the arrow UP, the arrow FR, and the arrow LH respectively

indicate an upper side in an up-down direction, a front side in a front-rear direction, and a left side in a left-right direction.

[0019] First, explanation follows regarding a configuration of the battery **10**. The battery **10** according to the present exemplary embodiment includes a laminated body **15**, primary sealing materials **33**, and a single secondary sealing material **40**.

[0020] The laminated body **15** includes plural battery cells **12**. The configuration of the laminated body **15** is well known, and therefore, the following explanation of the laminated body **15** is simplified. The laminated body **15** includes a single negative terminal electrode (electrode) **16**, a single positive terminal electrode (electrode) **20**, plural bipolar electrodes (electrodes) **25**, and plural separators **30**. Each bipolar electrode **25** is positioned between the negative terminal electrode **16** and the positive terminal electrode **20**. Each separator **30** is positioned between the positive terminal electrode **20**, the negative terminal electrode **16**, and the bipolar electrode **25** which are adjacent to each other. It should be noted that although the battery **10** that is illustrated in FIG. 2 to FIG. 4 includes two bipolar electrodes **25**, as long as there are plural bipolar electrodes **25**, any number may be used.

[0021] The negative terminal electrode **16** that configures a lower end portion of the laminated body **15** includes a current collector (current collector foil) **17**, a negative electrode active material layer **18** provided at one face of the current collector **17**, a conductive auxiliary agent, and a binder. The current collector **17** of the negative terminal electrode **16** is, for example, a nickel foil, the negative electrode active material layer **18** is natural graphite, and the binder is SBR/CMC. The positive terminal electrode **20** that configures an upper end portion of the laminated body **15** includes a current collector **17**, a positive electrode active material layer **21** provided at one face of the current collector **17**, a conductive auxiliary agent, and a binder. The current collector **17** of the positive terminal electrode **20** is, for example, an aluminum foil, the positive electrode active material layer **21** is an NCM-based active material, the conductive auxiliary agent is carbon black, and the binder is PVdF. Each bipolar electrode **25** that is positioned between the negative terminal electrode **16** and the positive terminal electrode **20** includes a current collector **17**, a negative electrode active material layer **18** provided at both sides of the current collector **17**, a conductive auxiliary agent at a negative electrode active material layer **18** side, a binder at the negative electrode active material layer **18** side, a positive electrode active material layer **21**, a conductive auxiliary agent at a positive electrode active material layer **21** side, and a binder at the positive electrode active material layer **21** side. The negative electrode active material layer **18** of the bipolar electrode **25**, the conductive auxiliary agent at the negative electrode active material layer **18** side, and the binding agent at the negative electrode active material layer **18** side are each the same as the negative electrode active material layer **18**, the conductive auxiliary agent, and the binding agent of the negative terminal electrode **16**, and the positive electrode active material layer **21** of the bipolar electrode **25**, the conductive auxiliary agent at the positive electrode active material layer **21** side, and the binding agent at the positive electrode active material layer **21** side are each the same as the positive electrode active material layer **21**, the conductive auxiliary agent, and the binding agent of the positive terminal electrode **20**. The laminated body **15** is configured by laminating the negative terminal electrode **16**, the positive terminal electrode **20**, the plural bipolar electrodes **25**, and the plural separators **30** in a laminating direction LD. In FIG. 1 and FIG. 2, the laminating direction LD is parallel to the up-down direction.

[0022] The planar shape of the separator **30** is a rectangle, and an outer peripheral portion of the separator **30** is positioned at a level lower compared to a portion other than the outer peripheral portion. The outer peripheral portion of the separator **30** is welded (spot welded) to the adjacent primary sealing material **33**. The separator **30** is a laminated body in which a sheet made of polyethylene (PE) is sandwiched between a pair of polypropylene (PP) sheets from above and below. The thickness of the separator **30** is, for example, 20 μm . The planar shapes of the negative terminal electrode **16**, the positive terminal electrode **20**, the bipolar electrodes **25**, and the

separators **30** of the present exemplary embodiment are rectangles. Therefore, the planar shape of the laminated body **15** is a rectangle. The respective separators **30**, and the negative electrode active material layer **18** and the positive electrode active material layer **21** positioned above and below the separators **30** are constituent elements of the battery cell **12**.

[0023] As illustrated in FIG. 2, a primary sealing material **33** made of a resin is provided at an outer peripheral portion of each current collector **17**, the outer peripheral portion of the current collector **17** being a region at which the negative electrode active material layer **18** and the positive electrode active material layer **21** are not formed. The primary sealing material **33** is a frame-like body with a planar shape that is a rectangle, and a groove is provided on an entire inner peripheral face of the primary sealing material **33**. The primary sealing material **33** is welded to the current collector **17** in a state in which the outer peripheral portion of the current collector **17** is inserted into the groove of the primary sealing material **33**.

[0024] When the laminated body **15** is configured as illustrated in FIG. 2, an outer peripheral portion of the laminated body **15** is surrounded by plural primary sealing materials **33**. A single secondary sealing material **40**, serving as a frame-like body having a rectangular shape in plan view, is molded at the outer peripheral portion of each primary sealing material **33** by injection molding that is performed in a state in which the laminated body **15** and the plural primary sealing materials **33** are positioned inside a molding die, which is not illustrated in the drawings. The type of resin material that serves as a constituent material of the secondary sealing material **40** is different from the type of resin material that serves as a constituent material of the primary sealing materials **33**. Plural communication holes (not illustrated in the drawings), which respectively communicate with a front end portion of an internal space V of each battery cell **12** in an airtight state and a liquid-tight state, are provided at a front portion of the secondary sealing material **40**.

[0025] Further, a liquid injection frame, which is not illustrated in the drawings, is integrally provided with the secondary sealing material **40** at a front end face of the secondary sealing material **40** during the above-described injection molding. The liquid injection frame is a hollow body, and a front end portion of the liquid injection frame is open. Plural communication holes (not illustrated in the drawings) that communicate with, in an airtight state and a liquid-tight state, the above-described respective communication holes are provided at a bottom plate portion (rear plate portion) of the liquid injection frame.

[0026] An electrolytic solution **45** (see FIG. 3) is filled into each internal space V via a front end opening portion of the liquid injection frame, an internal space, and the above-described communication holes, and the communication holes of the secondary sealing material **40**. After the electrolytic solution **45** has been filled, a front end face of the liquid injection frame is closed by a film-like member (not illustrated in the drawings) which is made of a resin.

[0027] The primary sealing materials **33**, the secondary sealing material **40**, and the liquid injection frame of the battery **10** having such a configuration are each configured by a resin material having insulation properties. The resin material of the primary sealing materials **33** is low density PE (polyethylene), and the melting temperature mp1 of the primary sealing materials **33** is from 100 to 115° C. The resin material of the secondary sealing material **40** is a different type of resin material than the low density PE. Further, the electrolytic solution **45** is, for example, a mixed solution of ethylene carbonate (EC), ethyl methyl carbonate (EMC), and/or dimethyl carbonate (DMC). The boiling point bp of the electrolytic solution **45** is a temperature between 10° and 230° C. However, the boiling point bp of the electrolytic solution **45** needs to be higher than the melting temperature mp1. Therefore, in a case in which the melting temperature mp1 is 115° C., for example, it is necessary to adjust the mixing ratio of ethylene carbonate (EC), the ethyl methyl carbonate (EMC), and/or the dimethyl carbonate (DMC) such that the boiling point bp of the electrolytic solution **45** is higher than 115° C. In the present exemplary embodiment, the mixing ratio of EC, EMC, and/or DMC is 1:1:1. It is further preferable that the boiling point bp of the electrolytic solution **45** is higher than the melting temperature mp1 by 50° C. or more.

[0028] A melting temperature smp of the separators **30** is higher than 160°C . The temperature of the battery **10** when the battery **10** is in a normal use state is defined as a normal battery operating temperature nth . The normal battery operating temperature nth is, for example, a temperature of less than 70°C . In the present exemplary embodiment, the following Expression (1) is satisfied.

$nth < mp1 < bp$ and $nth < mp1 < smp$ Expression (1):

[0029] The battery **10** stores (charges) when a voltage is applied from a DC power source.

Operation and Effects

[0030] Next, explanation follows regarding the operation and advantageous effects of the present exemplary embodiment.

[0031] Here, assume a case in which an electric vehicle that is installed with the battery **10** travels using force generated by an electric motor that is supplied with power from the battery **10**, and the battery **10** is not being charged or excessive charging has not been performed. The temperature of the battery **10** at this time is included in the normal battery operating temperature nth .

[0032] Next, assume a case in which, for example, voltage continues to be applied to the battery **10** despite the battery **10** being sufficiently charged, such that the battery **10** has increased in temperature to a temperature higher than the normal battery operating temperature nth .

[0033] In the present exemplary embodiment, Expression (1) is satisfied. Therefore, a state occurs in which the temperature of the battery **10** is higher than or equal to the melting temperature $mp1$ and lower than the boiling point bp when the temperature of the battery **10** has become high. In particular, in a case in which the boiling point bp is higher than the melting temperature $mp1$ by 50°C . or more, a state in which the temperature of the battery **10** is higher than or equal to the melting temperature $mp1$ and lower than the boiling point bp is easily maintained for a certain period of time. In such a case, as illustrated in FIG. 3, each primary sealing material **33** is melted. Therefore, the electrolytic solution **45** is capable of moving between the internal spaces V positioned above and below each current collector **17**, passing through the region in which the primary sealing material **33** was present. As a result, the negative electrode active material layer **18** that is provided at an upper face of each current collector **17** and the positive electrode active material layer **21** that is provided at a lower face of each current collector **17** are in liquid communication with each other by the electrolytic solution **45**. Namely, the negative electrode active material layer **18** that is provided at the upper face of each current collector **17** and the positive electrode active material layer **21** that is provided at the lower face of each current collector **17** are forcibly discharged, and the negative electrode active material layer **18** and the positive electrode active material layer **21** are equalized in potential. Therefore, even if each separator **30** is shut down due to the temperature of each separator **30** becoming high, the negative electrode active material layer **18** that is provided at the upper face of each current collector **17** and the positive electrode active material layer **21** that is provided at the lower face of each current collector **17** are forcibly discharged using the electrolytic solution **45**.

[0034] Thereafter, when the temperature of the battery **10** further increases, and the temperature of the battery **10** becomes higher than or equal to the melting temperature smp , the respective separators **30** are melted, as illustrated in FIG. 4. Therefore, the negative electrode active material layer **18** and the positive electrode active material layer **21** of the adjacent electrodes **16**, **20**, and **25** face each other in the up-down direction. As a result, the negative electrode active material layer **18** and the positive electrode active material layer **21** facing each other in the up-down direction may be surface short-circuited. However, even if a surface short circuit has occurred, the negative electrode active material layer **18** and the positive electrode active material layer **21** are already equalized in potential, such that there is little risk of the battery **10** becoming excessively high in temperature.

[0035] Although the battery **10** according to an exemplary embodiment has been explained as above, the battery **10** can be appropriately modified in design within a range that does not depart

from the gist of the present disclosure.

[0036] For example, the materials of the current collectors **17**, the negative electrode active material layer **18**, the positive electrode active material layer **21**, the separators **30**, the primary sealing materials **33**, the secondary sealing material **40**, and the electrolytic solution **45** may be different from those described above.

Claims

1. A bipolar battery, comprising: a plurality of bipolar electrodes each including a current collector, a negative electrode active material layer provided on one face of the current collector, and a positive electrode active material layer provided on another face of the current collector; a separator positioned between the negative electrode active material layer and the positive electrode active material layer, which are adjacent to each other; a plurality of frame-like primary sealing materials provided at an outer peripheral portion of the current collector, the outer peripheral portion being a region at which the negative electrode active material layer and the positive electrode active material layer are not provided; a single frame-like secondary sealing material provided at an outer peripheral portion of all of the primary sealing materials; and an electrolytic solution provided at an internal space surrounded by the current collector, the primary sealing materials, and the secondary sealing material, wherein, in a case in which a melting temperature of the primary sealing materials is defined as mp1, a normal battery operating temperature is defined as nth, a boiling point of the electrolytic solution is defined as bp, and a melting temperature of the separator is defined as smp, $nth < mp1 < bp$ and $nth < mp1 < smp$.
 2. The bipolar battery according to claim 1, wherein bp is higher than mp1 by 50° C. or more.
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